

Identifying Extreme [OIII] $\lambda 5007$ Emission-Line Galaxies in the Fields of Massive Clusters with VLT-MUSE

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ABSTRACT

The increasing number of JWST-JADES Ly α star-forming galaxy candidates, considered to be those driving early cosmic reionization, are limited to low-resolution rest-frame UV and visible observations; their dwarf [OIII]-extreme line emission (EELG) counterparts in the nearby Universe therefore provide important spectral analogues. These EELGs, however, present challenges to photometric selection due to their lack of stellar continuum: wide-band photometric surveys are therefore challenged by biases in redshift distributions. We instead present Integral Field (IFU) Spectroscopy as a direct selection method with VLT-MUSE, identifying a sample of [COMPLETE] EELGs in the fields of massive clusters. We present a source extraction pipeline and spectral fitting to infer the astrophysical properties of these galaxies, including obtaining metallicities and using legacy HST imaging to SED fit our pea candidates and infer their masses and star-forming rates. We further investigate the population distributions of these galaxies and compare to the MUSE flat-field, and new southern-sky populations of these galaxies.

Key words: galaxies:starburst – galaxies:dwarf

1 INTRODUCTION

One of the key ongoing questions in extragalactic astronomy involves understanding the process under which the intergalactic medium transitioned from opaque neutral hydrogen to the current transparent, ionised state: the epoch of reionization. It is accepted (Brunker et al. 2022) that the ionising radiation of this period $z > 6$ was comprised of that from star forming galaxies (Robertson et al. 2010; Tacchella 2025) and active galactic nuclei (Madau et al. 2004; Madau & Dickinson 2014); with the increasing number of highly star-forming galaxy candidates at redshifts above $z = 10$ discovered by the James Webb Space Telescope (JWST) (Rhoads et al. 2023; Hainline et al. 2023) it has become valuable to study their nearby counterparts, from which a more detailed picture can be built of their spectra.

It is believed the strongest contribution to ionizing radiation by SFGs is from extreme emission line galaxies (EELGs), a class of dwarf galaxies dominated by recent intense starburst, forming massive hot stars with UV-dominant emission (Tang et al. 2023; Davis et al. 2024). This strongly ionises the surrounding interstellar medium, causing fluorescence at specific atomic transitions, a process initially established by Menzel (1962). These extreme lines dominate the galaxies’ spectra, given the low-mass galaxies lack large stellar populations to contribute to the spectral continuum, and therefore give the observed atomic transition lines extreme equivalent widths (EWs; the fraction of integrated flux on a specific emission line to the continuum level); this process of starburst naturally produces large amounts of Lyman continuum (UV) radiation from the rapidly growing population of young, massive hot stars (Madau et al. 1999; Faucher-Giguère et al. 2008). It should be noted that this theory

of reionisation is complicated by the large neutral hydrogen column densities surrounding star forming nebulae in these young galaxies; Izotov et al. (2018) and Jaskot & Oey (2013) highlight that for low-mass galaxies a fully-ionized ISM or one with optically thin tunnels allow for the escape of UV radiation into the intergalactic medium. Recent observations of $z > 6$ galaxies have established strong correlation between [OIII]-H α emission and Lyman Continuum (UV) escape (Tang et al. 2023) (established initially at low z by Flury et al. (2022)), and similarly with UV luminosity (Endsley et al. 2024); in contrast with earlier (Leitherer & Heckman 1995) beliefs about starburst, deep-field astronomy has proven EELGs as direct probes of the (re-)ionising IGM.

These EELGs are distinguished by their dominant emission features, with the primary diagnostic depending on the galaxy’s redshift. For star-forming galaxies in the early universe these are rest-frame ultraviolet features: a prominent Lyman alpha emission line or break leads to ‘Ly- α -’ or ‘Ly-break-’ galaxy classification (Tacchella et al. 2023). Conversely, EELGs in the $z < 1$ universe are discerned by the strength of their [OIII] emission, a feature that is intrinsically visible and dominates their observed optical spectra: those nearby were discovered as a class by GalaxyZoo (Cardamone et al. 2009) identification from the Sloan Digital Sky Survey (SDSS), earning the name ‘Green Peas’ (GPs) due to their unresolved, [OIII]-extreme imaging causing them to appear green when artificially coloured.

As JWST-JADES has recently begun to discover more SFGs deep into the $z > 10$ EoR (Hainline et al. 2023), these GPs have become popular analogues of their high-redshift counterparts (e.g. Brunker et al. (2022); Izotov et al. (2018); Rhoads et al. (2023)): they too are low metallicity (Jiang et al. 2019), high specific star forming rate (sSFR) (Brunker et al. 2022; Liu et al. 2022), and of which a significant number (50%) show high Ly α escape fraction Jaskot

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& Oey (2013); Yang et al. (2017a). Detection of Lyman continuum leakage is rare at local redshifts ($z < 1$) and most of these local continuum-leaking galaxies are GPs (Yang et al. 2016; Izotov et al. 2016); those of the highest escape fraction are all Green Peas. This population of galaxies has therefore become a useful analogue of the SFG reionisation mechanism, and has therefore seen the recent development of large Pea catalogues (Samonski et al. 2025). More recently, these galaxies are often further classified into those with additional strong $H\alpha$ emission on $z < 0.36$ (Izotov et al. 2011) – ‘Purple Grapes’ – and extremely low-mass, very local $z < 0.1$ ‘Blueberry’ Galaxies (Yang et al. 2017b; Liu et al. 2022). For the purposes of this study we use the generalised Pea description; that of unresolved dwarfs with $\text{EW}[\text{OIII}] > 100$ in the nearby $z < 1$ universe.

We identify [OIII]-extreme (‘Pea-like’) galaxies from massive cluster observations by the Multi-Unit Spectroscopic Explorer (MUSE) on the Very Large Telescope (Bacon et al. 2010). This optical Integral Field Unit (IFU) instrument has a 1-arcminute² field of view on [4650, 9300] nm; this observation band makes the instrument an ideal choice for [OIII] selection on $z < 0.86$. These observations of galaxy clusters, the most massive gravitationally bound objects in the observable Universe, are part of a larger VLT-MUSE survey of cluster cores comparing strong lensing mass models to physical observations (Jauzac et al. 2019, 2021). This task is suited ideally to VLT-MUSE, which can detect galaxies to $z = 6$ (Bacon et al. 2015), and in this investigation allows for the identification of populations of foreground- cluster member- and background galaxies. Notably, because of their high mass, these clusters magnify background sources through gravitational lensing; this has become a powerful tool to map dark matter distributions in clusters (Massey et al. 2010), and can be used to magnify faint background galaxies at high z (prior to JWST this was the leading strategy for direct high- z detection using Hubble; see Livermore et al. (2017); Bouwens et al. (2017); Kawamata et al. (2018) for prominent examples).

Pea-like objects are often difficult to find uniformly with a traditional source finding procedure: their lack of stellar continuum often renders them invisible in photometric bands that do not align with specific emission lines and therefore creates biases in the populations scheduled for photometric follow-up. Recent efforts to instead use deeper spectroscopy and deeper image provide opportunities to establish estimates of the GP volume density (Amorín et al. 2015; Brunner et al. 2022; Kimsey-Miller et al. 2024), but are limited in census area, while SED-fitting-led wide-area methods instead are limited by lack of spectroscopic follow-up Yang et al. (2017b); Samonski et al. (2025).

In this paper we pose integral field spectroscopy (IFS) as a possible solution to these challenges. The technique has seen limited use before on individual GP observation (Bosch et al. 2019; Arroyo-Polonio et al. 2023) although it lacks any GP survey application. With the nearby advent of ELT-HARMONI, MUSE’s next-generation successor on the Extremely Large Telescope (Thatte et al. 2021), we showcase IFS as a key tool for Extreme Emission identification. From our observations we identify a uniform sample of EELGs, investigating their physical properties and comparing our populations with KISS and SDSS populations (Brunner et al. 2022; Samonski et al. 2025; Jiang et al. 2019). We further pose the identified sample in this paper as a useful calibration population for the large number pea-like objects detected by the Dark Energy Spectroscopic Instrument’s (DESI Collaboration et al. 2016) Survey (DESI).

Section 2: Observations outlines observational data and the source extraction pipeline used on candidate peas, including using corresponding HST photometry and a wide-field survey for population comparisons. We then outline population selection including

AGN contaminant rejection and line ratio tools to identify EELGs from the parent sample of candidates in **Section 3: EELG Identification**. From these identified peas, we investigate astrophysical properties in **Section 4: EELG Identification**, including investigations on SFR, mass and metallicity. Finally, we discuss extremes of this population and survey limits in comparison to contemporary work in **Section 5: Discussion**, concluding and outlining future work, including links to ELT-HARMONI in **Section 6: Conclusion**.

Distance-dependent quantities in this paper assume flat Λ CDM cosmology consistent with Planck Collaboration et al. (2020); $H_0 = 66.66 \text{ km s}^{-1}$ and $\Omega_m = 0.3097$.

2 OBSERVATIONS & DATA REDUCTION

2.1 VLT-MUSE and MUSE-WIDE

2.2 Hubble Photometry

3 EELG IDENTIFICATION

4 PHYSICAL PROPERTIES

4.1 Metallicity & SFR

4.2 Population Properties

5 DISCUSSION

6 CONCLUSION

7 CONCLUSIONS

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APPENDIX A: BLANK

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